Code No: 136FF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
INDUSTRIAL MANAGEMENT
(Common to CE, ME, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is the hierarchical nature of needs?	[2]
b)	State the social responsibilities of management.	[3]
c)	State the concept of organization chart.	[2]
d)	What do you understand by the term 'span of control' and 'unity of command'?	[3]
e)	What is value analysis? Explain in detail.	[2]
f)	Distinguish between cost and value.	[3]
g)	What are the different types of allowances to be considered in establishing standard time.	[2]
h)	Explain Ishikawa's Fishbone diagrams. What are its applications?	[3]
i)	State the advantages of job evaluation.	[2]
j)	Define the terms Job Description and Evaluation.	[3]

PART – B

(50 Marks)

2.a)	"Theory X and Theory Y are concerned with the nature of people". How does the job situation affect the application of this theory? What are its implications?	
b)	Critically examine Herzberg's two factor theory.	[5+5]

OR

3.a)	Explain briefly the Fayol's principles of management.	
b)	Differentiate between Japanese and American management with suitable examples.	[5+5]
4.a)	Distinguish between a military organization and a functional organization.	
b)	What is a network organization structure? Give the areas where it is best suitable for adopting. Give its strengths and weaknesses.	[4+6]

OR

5.a)	What is boundaryless organization? Give its chart, advantages and disadvantages.	
b)	What is its role in providing insights for designing organization structure?	[5+5]

6.a) Discuss the main objective which a factory planning engineer should attempt to achieve when designing a plant layout.

b) What is meant by a travel chart and show how such a chart can be of use in determining the best relative location of departments in a factory? [5+5]

OR

7. What are the pros and cons of using the location factor analysis method for location planning? Do you have any recommendations on how to use this method for location planning? [10]

8.a) A department store manager wishes to make a work sampling study to estimate the percentage time that the clerks are busy waiting for customers and percent time that they are idle. The current best guess is that clerks are idle 25 percent of the time. Determine the number of observation required if we wish to be 95 percent confident that the results is with in ± 1.5 percent, given number of observations at 20% is 2995 and at 30% it is 3750 for the same precision.

b) Compare stop watch study and work sampling in terms of cost to make studies, representatives of samples taken and comparative accuracy. [5+5]

OR

9.a) With reference to method study, describe the effect of the following factors:
i) Economic factors, ii) Technical factors iii) Human factors.

b) What are desirable characteristics of a sample taken for sampling plan? Explain. [5+5]

10.a) Explain the trade- off in network crashing.

b) Projects involve direct as well indirect costs and project managers need to use this information in project management. Comment on this statement. [5+5]

OR

11.a) Describe briefly the factor comparison method of job evaluation.

b) A project PERT network has only three possible paths, P-R, P-S-U, and Q-T-U. The activities, along with their expected time and standard deviations, are represented in ordered pairs as P(8, 2), Q(16, 4), R(15, 3), S(14, 2), T(5, 2), and U(5, 1). Develop the network and find the probability of completing the project in 25 days. [4+6]

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